Exp no:07 14.03.2025	SQL STRING FUNCTIONS

AIM:

To perform sql string operations by using string functions.

STRING OPERATIONS USING DUAL TABLE:

1. **ASCII(CHR):**
   To find the ascii value of the character

```
SQL> select ascii('Y') as Asciivalue from dual;

ASCIIVALUE
----------
        89
```

2. **CHR(NUMBER):**

To find the character of the given number

```
SQL>  select chr(80) as character from dual;

 C
 -
 P
```

3 .CONCAT(string1,string2):

To concatenate to two strings.

```
SQL> select concat('YOKA','RAJAN') as concated_string from dual;
CONCATED_
---------
YOKARAJAN
```

4.UPPER(STR):

Convert the string into uppercase.

```
SQL> select upper('yoks') as caps from dual;


CAPS
----
YOKS
```

5.LOWER(STR):

Convert the string into lowercase.

```
SQL> select lower('YOKARAJAN') as caps from dual;


CAPS
---------
Yokarajan
```

6. LENGTH(STR):

It finds the length of the given string.

```
SQL> select length('*My name is Yokarajan*') as length from dual;
    LENGTH
----------
        22
```

7. TRIM(STR):

Removes the extra white spaces in the given string

```
SQL> select trim(' My name is Yokarajan ') as trimmed_string from
dual;
```

```
TRIMMED_STRING
--------------------
My name is Yokarajan
```

```
SQL> select ltrim('  Yoks   ') as left_trim from dual;
LEFT_TR
-------
Yoks
```

```
SQL> select rtrim('  Yoks   ') as right_trim from dual;
RIGHT_
------
  Yoks
```

8. REPLACE():

Replaces the given string instead of the available one.

```
SQL> select replace('Kongu Engineering College','College','Clg') as
replaced_string from dual;
REPLACED_STRING
---------------------
Kongu Engineering Clg
```

9. SUBSTR():

Finds the substring within the given range.

```
SQL> select substr('I am Yoks', 6, 4) as substring from dual;
SUBS
----
Yoks
```

10. RPAD():

Removes the characters from right position of the string with specified positions.

```
SQL> select rpad('Yokarajan',7) as RPAD from dual;
RPAD
-------
Yokaraj

SQL> select rpad('Yokarajan',12,'*') from dual;
RPAD('YOKARA
------------
Yokarajan***
```

<u>11.LPAD:</u>

```
SQL> select lpad('Yokarajan',12,'*') from dual;
LPAD('YOKARA
------------
***Yokarajan
```

<u>TABLE:</u>

```
SQL> select * from staff;
        ID NAME       DEPT     SALARY        AGE
---------- ---------- ---- ---------- ----------
       101 Arun       CSE       50000         35
       102 Bala       ECE       60000         40
       103 Yokarajan  IT        55000         27
       104 Ram        IT        58000         42
       105 Raju       EEE       40000         45
       106 Dharun     CHEM      35000         24
```

1. CONCATENATION:

```
SQL> select name || ' is a staff of ' || dept as department_info
from staff where salary > 40000;


DEPARTMENT_INFO

----------------------------

Arun is a staff of CSE

Bala is a staff of ECE

Yokarajan is a staff of IT

Ram is a staff of IT
```

2. UPPER:

```
SQL> select upper(name) as names from staff where name like '____';

NAMES

----------

ARUN

BALA

RAJU
```

3.LOWER:

```
SQL> select lower(name) as names, lower(dept) as department from
staff where dept in (select dept from staff group by dept having
avg(salary) > 30000);

NAMES      DEPA

---------- ----

ram        it

yokarajan  it

raju       eee

arun       cse

dharun     chem

bala       ece
```

4.Length:

```
SQL> select length(name) as length from staff where name in (select name from staff where salary > 50000);

    LENGTH
----------
         4
         9
         3
```

5.TRIM:

```
SQL> select trim(name) as trimmed_name from staff where id = 103 or id = 106;

TRIMMED_NA
----------
Yokarajan
Dharun
```

6. REPLACE:

```
SQL> select replace(name, 'Yokarajan', 'Yoks') as replaced_name from staff where id=103;

REPLACED_NAME
----------------------------------------
Yoks
```

7.RIGHT PAD:

```
SQL> select rpad(name, 7, '1') as rpad_name from staff where id=103;

RPAD_NA
-------
Yokaraj
```

8.LEFT PAD:

```
SQL> select lpad(name, 7, '1') as lpad_name from staff where id=103;


LPAD_NA
-------
Yokaraj
```

9. INSTR:

```
SQL> select instr(name, 'Yokarajan') as pos from staff;

       POS
----------
         0
         0
         1
         0
         0
         0

6 rows selected.
```

8.REVERSE:

```
SQL> select reverse(name) as reversed_string from staff where name
like '_%n';

REVERSED_S
----------
nurA
najarakoY
nurahD
```

STRING OPERATIONS RELATED TO EMAIL – REGEX:

```
SQL> select * from email;

        ID NAME       EMAIL
---------- ---------- --------------------
       101 Arun       arun@kongu.edu
       102 Bala       bala@kongu.edu
       103 Yokarajan  yokarajan@kongu.edu
       104 Ram        ram@kongu.edu
       105 Raju       raju@kongu.edu
       106 Dharun     dharun@kongu.edu
       107 Viji
       108 Rose
8 rows selected.
```

1. EXTRACTING THE DOMAIN NAME OF EMAIL:

```
SQL> select substr(email, instr(email, '@') + 1) as email_domain
from email;

EMAIL_DOMAIN
--------------------
kongu.edu
kongu.edu
kongu.edu
kongu.edu
kongu.edu
kongu.edu
8 rows selected.
```

2. CHANGING DOMAIN NAME:

```
SQL> select replace(email, substr(email, instr(email, '@')),
'@gmail.com') as new_mail from email;
```

```
NEW_MAIL
--------------------------------------------------------------------
------------
arun@gmail.com
bala@gmail.com
yokarajan@gmail.com
ram@gmail.com
raju@gmail.com
dharun@gmail.com
```

SEARCHING OPERATIONS USING LIKE AND NOT LIKE KEYWORD:

```
SQL> select * from email where name like 'R%';

        ID NAME       EMAIL
---------- ---------- --------------------
       104 Ram        ram@kongu.edu
       105 Raju       raju@kongu.edu
       108 Rose

SQL> select * from email where name not like 'R%';

        ID NAME       EMAIL
---------- ---------- --------------------
       101 Arun       arun@kongu.edu
       102 Bala       bala@kongu.edu
       103 Yokarajan  yokarajan@kongu.edu
       106 Dharun     dharun@kongu.edu
       107 Viji
```

```
SQL> select email from email where email like '____@%';

EMAIL
--------------------
arun@kongu.edu
bala@kongu.edu
raju@kongu.edu


SQL> select count(*) as null_id from email where email is null;

   NULL_ID
----------
         2

SQL> select email from email where email is not null;

EMAIL
--------------------
arun@kongu.edu
bala@kongu.edu
yokarajan@kongu.edu
ram@kongu.edu
raju@kongu.edu
dharun@kongu.edu
6 rows selected.

SQL> select count(email) from email;

COUNT(EMAIL)
------------
           6
```

CONTENTS	MARKS ALLOTED	MARKS OBTAINED
Aim, Algorithm, SQL, PL/SQL	30	
Execution and Result	20	
Viva	10	
Total	60	

RESULT:

Thus different types of string operations using SQL string functions were executed